Y1 Block B Final Presentation

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Y1, Block B, Final Presentation

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Project Description

- Medical data
- Lots of correlations (EDA)
- Lots of options

Problem overview

- Regression

- Age

Model overview

Regression

- Linear Regression
- Decision Tree
- Gradient Boosting
- XGBoost

- Load, pre-process, split

Classification

- Logistic Regression
- Random Forest
- K-Nearest Neighbors

- Load, pre-process, split

Clustering

- K-Means
- Random Hierarchical
- DBSCAN

- Load, pre-process, split, reduce dimensionalities

Trade off

Performance

- Hyperparameter tuning
- Models
- Computation

Interpretability

- Outside your knowledge
- At a glance
- Crash

Analysation & justification

Regression

- Decision Tree
- Eeks out win
- Lowest scores

Classification

- Random Forest
- Convincing
- Accuracy...

Clustering

- Random Hierarchical
- Very convincing
- High score

Final analysis

- Narrow difference between regression
- Classification: best, not good
- RHC clear winner in clustering